

Krishnaa Vadivel

Scientific Machine Learning | Computational Materials

New Haven, CT | srikrishnaa.careers@gmail.com | 516-853-5663

Portfolio: <https://krishnaa423.github.io/personal-website/> | GitHub: <https://github.com/krishnaa423>

LinkedIn: <https://www.linkedin.com/in/sri-krishnaa-ece-phd/>

Summary

Scientific-ML researcher with a physics-first background in equivariant graph networks, automatic differentiation, first-principles modeling, and GPU/HPC implementation.

Technical Skills

ML: JAX, PyTorch, PyTorch Geometric, e3nn, equivariant GNNs, automatic differentiation, NumPy, SciPy

Scientific compute: Python, C++, MPI, OpenMP, CUDA (cuBLAS, cuTENSOR), ROCm (rocBLAS, rocFFT), PETSc, SLEPc, DOLFINx

Platforms: Frontier, Perlmutter, Frontera, Docker, Kubernetes, Linux

Experience

PhD Researcher

Yale University, Electrical Engineering

2021–present

- Build DeePH-style computational graphs for LiF and double-perovskite systems; train equivariant models in PyTorch Geometric and use automatic differentiation to obtain force and optical-property quantities.
- Connect ML models to first-principles exciton and many-body theory, ensuring datasets, losses, and evaluations map to physical observables.
- Build and debug distributed GPU workflows in Python/C++ using MPI, OpenMP, CUDA/ROCm, PETSc, and SLEPc on national HPC systems.
- Prototype retrieval-driven scientific assistants that extract structured simulation inputs from papers and technical documentation.

Software and Embedded Systems Engineer

Probe Technologies / Weatherford Wireline Services, Houston, TX

2019–2021

- Built CUDA-enabled waveform-processing and visualization tools, adding production experience with data pipelines and performance constraints.
- Implemented SQL-backed ASP.NET services and C# tools for remote engineering-test workflows.

Selected Projects

EGNN QM9 force model — Trained an equivariant GNN and differentiated predicted molecular energies to obtain forces.

Materials optical-property models — Applied graph networks and automatic differentiation to accelerate costly calculation stages; roughly 100x speedup in parts of the workflow.

Scientific retrieval pipeline — Structured software documentation and paper content into starter artifacts for computational-science workflows.

Education

PhD, Electrical Engineering

Yale University, Computational materials and scientific ML

2021–present

MPhil and MS, Electrical Engineering

Yale University

2023

MEng and BS, Electrical and Computer Engineering

Cornell University,

2013–2018

Research Record

Co-author, *Journal of the American Chemical Society*, 2025; APS presentations, 2024–2026.